

STUDENT BOOK ANSWERS

Chapter 4 Chemical reactions

Worked example 4.1

- a** Iron + oxygen $\rightarrow$ iron oxide
b Octane + oxygen $\rightarrow$ carbon dioxide + water

Worked example 4.2

- a** $4\text{Fe(s)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{Fe}_2\text{O}_3\text{(s)}$
b $\text{C}_6\text{H}_{12}\text{O}_6\text{(l)} + 6\text{O}_2\text{(g)} \rightarrow 6\text{H}_2\text{O(g)} + 6\text{CO}_2\text{(g)}$

Question set 4.1

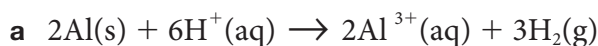
- 1 A chemical equation represents a chemical reaction.
- 2 Write the reactants and products using their chemical formula. Start with the elements that only appear in one substance on each side of the equation. Only add numbers in front of the formula. Check each element. Add the states.
- 3 **a** $\text{Fe(s)} + 2\text{HCl(aq)} \rightarrow \text{FeCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
b $2\text{NaHCO}_3\text{(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{Na}_2\text{SO}_4\text{(aq)} + 2\text{CO}_2\text{(g)} + 2\text{H}_2\text{O(l)}$
c $2\text{CuFeS}_2\text{(s)} + 2\text{SiO}_2\text{(s)} + 4\text{O}_2\text{(g)} \rightarrow \text{Cu}_2\text{S(s)} + 2\text{FeSiO}_3\text{(s)} + 3\text{SO}_2\text{(g)}$
d $2\text{PbS(s)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{PbO(s)} + 2\text{SO}_2\text{(g)}$
e $2\text{C}_3\text{H}_8\text{(g)} + 10\text{O}_2\text{(g)} \rightarrow 6\text{CO}_2\text{(g)} + 8\text{H}_2\text{O(l)}$
- 4 **a** No: $2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{H}_2\text{O(l)}$
b Yes
c No: $\text{C}_6\text{H}_{12}\text{O}_6\text{(aq)} \rightarrow 2\text{C}_2\text{H}_5\text{OH(aq)} + 2\text{CO}_2\text{(g)}$
d Yes
e Yes
- 5 **a** $\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightarrow 2\text{NH}_3\text{(g)}$
b $\text{Mg(OH)}_2\text{(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + 2\text{H}_2\text{O(l)}$
c $\text{Zn(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{H}_2\text{(g)}$

Experiment 4.2: Reactivity of metals

Discussion

Calcium metal should have been the most reactive metal. It reacts easily with water.
Copper is the least reactive metal as the reaction with 4 M HCl was slow.

Worked example 4.3



Question set 4.2

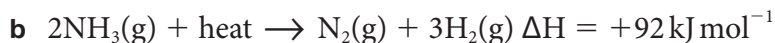
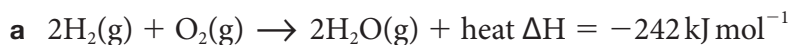
- 1 **a** Spontaneous chemical reactions are reactions that occur at room temperature without the input of energy such as electricity or heat.
b Spectator ions are species on both sides of an ionic equation that do not take part in the reaction.
- 2 Balanced chemical equations and net ionic equations.
- 3 Reactivity is a chemical property as the arrangement of the atoms change in the reaction.
- 4 **a** $\text{Fe(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Cu(s)} + \text{Fe}^{2+}(\text{aq})$
b $\text{CO}_3^{2-}(\text{aq}) + \text{Mg}^{2+}(\text{aq}) \rightarrow \text{MgCO}_3(\text{s})$
- 5 **a** $\text{Ca(s)} + \text{H}_2\text{O(l)} \rightarrow \text{Ca}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) + \text{H}_2(\text{g})$
b $\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2(\text{g})$
c $\text{Zn(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$
d $\text{Pb}^{2+}(\text{aq}) + 2\text{I}^-(\text{aq}) \rightarrow \text{PbI}_2(\text{s})$

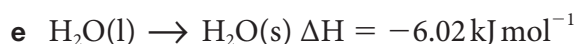
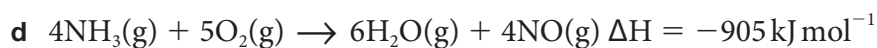
Experiment 4.3: Endothermic and exothermic reactions

Discussion

- 1 Exothermic reactions gave off heat and the temperature rose. Endothermic reactions absorbed heat and the temperature decreased.
- 2 $\text{NaOH(s)} + \text{water} \rightarrow \text{NaOH(aq)} + \text{heat}$
 $\text{KNO}_3(\text{s}) + \text{water} + \text{energy} \rightarrow \text{KNO}_3(\text{aq})$
 $2\text{CH}_3\text{COOH(aq)} + \text{Na}_2\text{CO}_3(\text{s}) \rightarrow 2\text{CH}_3\text{COONa(aq)} + \text{CO}_2(\text{g}) + \text{H}_2\text{O(l)} + \text{heat}$
- 3 Styrofoam cups insulated the reactions so heat was less likely to be lost to or absorbed from the environment.

Worked example 4.4





Worked example 4.5

1 a 298 K

b 269 K

2 a 362°C

b -135°C

Experiment 4.4: Comparing heat capacities

Analysis of results

Beaker B had the greater temperature change in the first experiment with the water.

Discussion

Water has a higher specific heat capacity than vegetable oil. To heat the water to 50°C would mean more heat given to the beaker. When added to beaker B there is more heat energy available to heat beaker B and so it has the greatest temperature change. If vegetable oil had the greater specific heat capacity then the reverse would be true.

Worked example 4.6

a $q = mC\Delta T, q = 15 \times 2.46 \times 40 = 1476 \text{ J}$

b $q = mC\Delta T, m = \frac{q}{C \Delta T} = \frac{2346}{2.00 \times 30} = 39.1 \text{ g}$

Question set 4.3

1 a Endothermic: When energy is absorbed from the surroundings.

b Specific heat capacity: The heat needed to increase the temperature of 1 g of the substance by 1 K.

2 Temperature ($^\circ\text{C}$) = temperature (K) -273



4 283.2 K

5 As tin has a smaller specific heat capacity than Al, it would increase in temperature more quickly than the aluminium with the same rate of energy input. As $q = mC\Delta T$, if the same energy (q) has reached both in the short term, ΔT for the tin would be larger because C is smaller. After 2 hours, however, both pieces would be the same temperature but the aluminium would have absorbed more heat energy to get there.

6 $q = mC\Delta T = 500 \times 2.00 \times 70 = 70\,000 \text{ J}$

7 The student should put the water into the copper beaker as it takes less to heat copper than the pyrex glass. The specific heat capacity of copper is lower than that of Pyrex. Also, copper is a better conductor of heat so it will transfer heat to the contents more efficiently.

Worked example 4.7

$q = mC\Delta T$, $q = 100 \times 4.18 \times 30 = 12\,540\text{ J}$. This was for 0.372 g. For 1 g, $12\,540/0.372 = 33\,709\text{ kJ}$ for 1 g of 1-propanol, assuming all the heat from the combustion went into the flask's contents.

Question set 4.4

- 1 a Fossil fuels: Fuels from decayed remains of plants and animals produced over millions of years, such as crude oil and coal
b Law of conservation of energy: Energy is not created or destroyed; it can only be changed from one form to another.
- 2 Complete combustion of hydrocarbons will produce carbon dioxide and water. Incomplete combustion of hydrocarbons means the products will contain water, carbon and carbon monoxide. The ratio of carbon and carbon monoxide will depend on the oxygen level present.
- 3 Incomplete combustion produces products (CO, C) that have more energy than carbon dioxide. The change in energy between the reactants and products is not as great in incomplete combustion reactions. This is why they have a lower heat of combustion.
- 4 a $\text{C}_6\text{H}_{12}(\text{l}) + 9\text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$
b $2\text{C}_2\text{H}_6(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{CO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$
or $2\text{C}_2\text{H}_6(\text{g}) + 3\text{O}_2(\text{g}) \rightarrow 4\text{C}(\text{s}) + 6\text{H}_2\text{O}(\text{g})$
- 5 It is incomplete combustion as carbon monoxide has been produced. More oxygen is required to convert the carbon monoxide to carbon dioxide.

Chapter review questions

- 1 a Net ionic equation: The equation without spectator ions
b Exothermic: Energy is given off
c Enthalpy: Heat absorbed or given off in the reaction
d Renewable fuel: A fuel that can be produced at a faster rate than it is consumed
- 2 Many answers such as: biodiesel, gas, coal, petroleum
- 3 a Water molecules have to lose energy: exothermic
b Energy required: endothermic
c Change in enthalpy is negative: exothermic
d Energy is absorbed from the sunlight: endothermic
- 4 The two have slightly different chemical compositions.

	Temperature	
	(°C)	(K)
Melting point of mercury	− 39	234
Freezing point of helium	− 272	1
Boiling point of ethanol	78	351
Melting point of sodium chloride	801	1074

- 6 a $4\text{FeAsS(s)} + 11\text{O}_2\text{(g)} + 6\text{H}_2\text{O(l)} \rightarrow 4\text{FeSO}_4\text{(aq)} + 4\text{H}_3\text{AsO}_3\text{(aq)}$
 b $\text{C}_3\text{H}_8\text{(g)} + 5\text{O}_2\text{(g)} \rightarrow 3\text{CO}_2\text{(g)} + 4\text{H}_2\text{O(g)} \Delta H = -2220 \text{ kJ mol}^{-1}$
- 7 a $\text{Fe}_2\text{O}_3\text{(s)} + 3\text{H}_2\text{(g)} \rightarrow 2\text{Fe(s)} + 3\text{H}_2\text{O(l)}$
 b $2\text{Al(s)} + 3\text{Cl}_2\text{(g)} \rightarrow 2\text{AlCl}_3\text{(s)}$
 c $\text{Cu(s)} + 2\text{AgNO}_3\text{(aq)} \rightarrow 2\text{Ag(s)} + \text{Cu(NO}_3)_2\text{(aq)}$
 d $\text{V}_2\text{O}_5\text{(s)} + 5\text{Ca(s)} \rightarrow 5\text{CaO(s)} + 2\text{V(s)}$
 e $\text{H}_2\text{O}_2\text{(l)} \rightarrow 2\text{H}_2\text{O(l)} + \text{O}_2\text{(g)}$
- 8 A: Solid, melting point of 1768°C is above 25°C
 B: Gas, melts at -102°C and boils at -34°C
 C: Solid, melts at 57°C
 D: Liquid, melts at -8°C and boils at 59°C
 E: Solid, melts at 29°C , which is above 25°C
- 9 a The styrofoam cup is an excellent insulator, so the rate of heat transfer from the contents is very slow. Also, the mass of styrofoam is very small so heating it up will take little energy compared to a pyrex beaker, even if it has a bigger specific heat capacity.
 b Copper should be used as the experiment is trying to calculate the heat given off from the combustion of the spirit burner fuel. As copper has the lowest specific heat capacity and it is a superior heat conductor, so it takes less of the heat to warm the container before transferring heat to the contents, and the heat can be transferred more quickly preventing losses to the environment. It is not 'lost' by heating up the material so much, or in hot gases from the flame not losing their heat to the container.
- 10 a $q = mC\Delta T$, $q = 150 \times 4.18 \times (39.6 - 16.3) = 14\,609 \text{ J}$. The heat of combustion (kJ/g) = $14.609 \text{ kJ} / 0.48 \text{ g} = 30.44 \text{ kJ g}^{-1}$
 b Heat would have been lost to the surroundings and in heating up the water container. This would reduce the heat available to heat the water resulting in a lower temperature change.
- 11 Students' answers will vary.
- 12 There are slight differences in the chemical properties of diesel and biodiesel. Diesel has a slightly higher heat of combustion. Biodiesel's heat of combustion will vary with different crops. This makes diesel slightly more energy efficient than biodiesel and a more consistent fuel to use. Biodiesel has lower net CO_2 and CO emissions than diesel but slightly higher emissions of nitrogen oxides.
 Biodiesel is a renewable fuel and can be produced to meet the demands of society. However it requires a feedstock material, such as peanut oil, canola oil or soybean oil, which are food crops. Diesel is a fossil fuel and the world's reserves of fossil fuels are being consumed much quicker than they can be produced.
 Biodiesel is relatively carbon neutral as the carbon dioxide produced when it is burnt has been recently absorbed from the atmosphere by photosynthesis in plants. Some fossil fuels are probably used in the production and processing of the plant material. Diesel is not carbon neutral as it releases the carbon from the fossil fuels that were buried millions and millions of years ago.
- 13 a Students' answers will vary: member of the public will not understand the symbols used so can understand the worded equations.
 b Students' answers will vary: chemical equations provide more detailed information than word equations.
- 14 Students' answers will vary.